L laboratory instruments

P Online refractometer

S Software

A Automation system

SCHMIDT HAENSCH

自公元1864-160年光电检测仪器制造经验

德国S+H

Coloromat100-V2

Spectrophotometer

Professional white sugar color value measurement, you can directly use the 100mm funnel sample tube to pour the sample

parameters	Coloromat100 – V2
Measurement principle	Principles of single beam spectrophotometry
Wavelength range	340 – 900 nm*
Preset wavelength	560, 420, 720 nm
FWHM full width at half maximum	420 nm = 4 nm** 720 and 560 nm = 7 nm**
Wavelength operation	touch screen
light source	LED
Measuring range	0 – 16,000 ICUMSA Units (IU)
Accuracy	± 0.020 Extinction 420 nm ± 0.010 Extinction 560 nm ± 0.010 Extinction 720 nm ± 0.020 Extinction Other Wavelengths
Extinction accuracy	0.001
ICUMSA resolution	1 IU
IU accuracy	± 10 % for colors < 20 IU ± 5 % for colors > 20 IU
Operation interface	Touch screen operation Various methods built-in
Data display	Graphical interface
data output	RS232C & USB
operating temperature	10 – 40°C
language	English and German / Indonesian / Russian / Spanish

* For WL < 500 nm and > 700 nm, additional LED's may be required to ensure measurement accuracy and stability

** This very narrow half-value width enables high measurement accuracy as well as very good repeatability

- Measurements are reliable and reproducible
- 3 preset wavelengths 560, 420 and 720nm
- Extremely narrow peak width
- 6 selectable wavelengths (in the range 340-900 nm)
- Automatic wavelength switching
- Optical sample tubes up to 100mm long can be used for direct pouring injection or conventional cuvettes
- Single or continuous monitoring measurements
- Remote control via PC
- Data can be output to a computer or directly connected to a printer, or via USB.
- Up to 250 programmable methods
- GLP/GMP compliant documents
- Maintenance free

Application

- Determination of color value of white sugar (ICUMSA determination method GS2/3-10), national standard method
- Determination of beer and wort color (EBC color number)
- Determination of color index of soft drinks and red wine
- Simple extinction value determination
- Color determination of extracts such as coffee
- Determination of APHA value in color determination
- Monitor color changes over time based on functional design requirements
- Enzymatic assays such as NAD(P)- and NAD(P)H-methods

Single-beam spectrophotometry is used to determine the color value of liquids.

Coloromat100 spectrophotometer can be very conveniently used for transmission, extinction and color value measurement;
You can use a polarimetric tube up to 100mm as a sample tube, pour the sample directly into it, or use a conventional ratio color pan measurement;
The sample chamber opens upward and liquid will not enter the instrument;
Adopting a unique optical design, the measurement is not affected by light scattering and complies with GLP/GMP requirements;
Continuous detection mode can be used to directly measure the color change of flowing samples and construct a function;
The instrument operating software can be upgraded via an online PC, and the instrument itself can pre-store up to 250 different test methods
, measurement of up to 99 samples;
Results can be stored directly in the instrument's internal memory;
Batch numbers, user information, etc. can be quickly entered through the touch screen;
Transmittance, extinction and color units can be easily measured in the range between 340 and 900 nanometers. for
For wavelengths below 500nm and above 750nm, additional LEDs may be required to ensure accurate measurements.
accuracy and stability.
Wavelengths are selected via the touch screen, with three wavelengths pre-installed as standard, the Coloromat 100 can also be equipped with 6 additional customized wavelengths;

When using a 100mm polarimetric tube as the sample tube, the sample can be injected quickly and easily by pouring, without the need for traditional
When measuring cuvettes, they need to be cleaned frequently. The measurement accuracy and resolution of the 100mm sample tube is that of the traditional 10mm
10 times the size of the cuvette. Compared with the 10mm cuvette, the measurement accuracy and resolution are increased by 10 times.

SCHMIDT HAENSCH
innovators by tradition since 1864

ISO 9001

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